

DISCRETE MATHEMATICS

LECTURE 11

0.1. Linear codes.

Definition 0.1. A code $C \subseteq F^n$ is called *linear* if it is a linear subspace of F^n . If $\dim(C) = m$, we call it an $[n, m]$ code.

Example 0.2. The previous example (of a perfect code) is the subspace generated by (1011) and (0112) in $\mathbb{Z}_3^4$. It is a linear $[4, 2]$ code.

For linear codes the separation is easier to compute.

Definition 0.3. The *weight* $w(\mathbf{x})$ of a code word $\mathbf{x} = (x_1, \dots, x_n)$ is the number of non-zero coordinates and the *weight of a linear code* C is $w(C) = \min\{w(\mathbf{x}) \mid \mathbf{x} \in C, \mathbf{x} \neq \mathbf{0}\}$.

Theorem 0.4. For a linear code $w(C) = d(C)$.

Proof. A linear code that contains the two words $\mathbf{x}$ and $\mathbf{y}$ also contains their difference $\mathbf{x} - \mathbf{y}$. The result follows from the fact that the Hamming distance $d(\mathbf{x}, \mathbf{y})$ is equal to the weight $w(\mathbf{x} - \mathbf{y})$.

An alternative proof is the following.

Assume that $\mathbf{x}, \mathbf{y} \in C$ are code words with minimal distance $d(C) = d(\mathbf{x}, \mathbf{y})$. Then $\mathbf{x} - \mathbf{y} \in C$ because C is a linear code. Then $w(C) \leq w(\mathbf{x} - \mathbf{y}) = d(\mathbf{x}, \mathbf{y})$, and so $w(C) \leq d(C)$.

Now assume that $\mathbf{z} \in C$ is of minimal weight, so $w(C) = w(\mathbf{z}) = d(\mathbf{z}, \mathbf{0}) \geq d(C)$. (Recall that $\mathbf{0} \in C$ since C is linear.)

Hence we have $d(C) \leq w(C) \leq d(C)$, which yields the desired equality. $\square$

Example 0.5. The linear code in $\mathbb{Z}_3^4$ generated by (1011) and (0112) has elements of the form

$$a(1011) + b(0112) \quad \text{for } 0 \leq a, b \leq 2.$$

We can list all nine elements as follows:

$$\begin{aligned} a = 0 : & \quad (0000) \quad (0112) \quad (0221), \\ a = 1 : & \quad (1011) \quad (1120) \quad (1202), \\ a = 2 : & \quad (2022) \quad (2201) \quad (2210). \end{aligned}$$

The non-zero elements all have weight 3, which gives that $w(C) = 3$. This confirms that the separation is 3 as we have computed before.

For linear codes we need to look at the M elements of the code instead of $\binom{M}{2} = \frac{M(M-1)}{2}$ pairs of elements to find the separation.

Definition 0.6. A *generator matrix* for a linear $[n, m]$ code in F^n is a $m \times n$ matrix such that its rows form a basis for C .

Example 0.7. Consider $C \subseteq \mathbb{Z}_3^4$ generated by (1112), (1001) and (2120). The linear code C has the following generator matrix:

$$G = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 0 \end{pmatrix}.$$

Let us simplify G using addition of a multiple of one row to another row. And multiplying a row by a non-zero constant. This will obviously not change the code generated by G .

$$\begin{aligned} G = \begin{pmatrix} 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 0 \end{pmatrix} &\xrightarrow[r_3 - 2r_1]{r_2 - r_1} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 2 & 2 \\ 0 & 2 & 0 & 2 \end{pmatrix} \\ &\xrightarrow[r_3 - r_2]{} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 1 & 0 \end{pmatrix} &\xrightarrow[2r_2]{} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\ &\xrightarrow[r_1 - r_3]{r_2 - r_3} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\ &\xrightarrow[r_1 - r_2]{} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} =: \tilde{G}. \end{aligned}$$

Definition 0.8. A generator matrix of the form $[I_m \mid A]$ is said to be of *normal form*.

The code words in the above example are

$$(x_1 \ x_2 \ x_3) \tilde{G} = (x_1 \ x_2 \ x_3 \ (x_1 + x_2)).$$

Sometimes we need to swap two columns to get to the normal form.

Example 0.9. (Part handout) The code C is given by

$$\begin{aligned}
 G = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 2 \end{pmatrix} &\xrightarrow[r_3 - 2r_1]{r_2 - r_1} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 2 & 2 & 0 \end{pmatrix} \\
 &\xrightarrow{r_3 - r_2} \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{cannot continue unless we swap } c_3 \text{ and } c_4 \\
 &\longrightarrow \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\
 &\xrightarrow{2r_2} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \text{(Handout)} \\
 &\xrightarrow[r_1 - r_3]{r_2 - r_3} \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\
 &\xrightarrow{r_1 - r_2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.
 \end{aligned}$$

The matrix

$$G'' = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

of normal form generates $C' = \{(x_1 \ x_2 \ x_3)G'' = (x_1 \ x_2 \ x_3 \ (x_1 + x_2)) \mid x_1, x_2, x_3 \in \mathbb{Z}_3\}$. The code C' is called *equivalent* to the original code $C = \{(x_1 \ x_2 \ (x_1 + x_2) \ x_3) \mid x_1, x_2, x_3 \in \mathbb{Z}_3\}$. Words in C' are obtained by swapping the third and fourth letters in a word in C .

Theorem 0.10. Every linear $[n, m]$ code C is equivalent to a code with generator matrix of the form $[I_m \mid A]$.

0.2. Dual codes and control matrices. We want to find a way to detect and correct errors in a received message that works for linear codes. To do that, we need the concept of a dual code.

Definition 0.11. The *dual code* $C^\perp$ of a linear code C in F^n is the linear code

$$C^\perp = \{\mathbf{y} \in F^n \mid \langle \mathbf{x}, \mathbf{y} \rangle = 0 \text{ for all } \mathbf{x} \in C\}.$$

Here $\langle, \rangle$ denotes the scalar product $\langle \mathbf{x}, \mathbf{y} \rangle = x_1y_1 + \cdots + x_ny_n$.

Looking at a generator matrix G for C , we see that $C^\perp$ is the nullspace and C is the row space. Thus by the dimension theorem, we have $\dim C^\perp = n - \dim C = n - m$.

Remark 0.12. (Handout) In $\mathbb{R}^n$ any vector $\mathbf{u}$ can be uniquely decomposed into $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$ where $\mathbf{u}_1 \in U$ and $\mathbf{u}_2 \in U^\perp$ for any subspace U . This is not true for general F^n . (The problem is to construct an orthonormal basis for U . For example for $U = \langle (1 \ 1 \ 1) \rangle$ in $\mathbb{Z}_3^3$.)

Example 0.13. Sometimes we even have $C = C^\perp$, a so-called *self-dual code*. One example is

$$G = \begin{pmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{pmatrix}$$

in $\mathbb{Z}_5^4$. The dual code is 2-dimensional and contains the rows in G as they are perpendicular to each other and to themselves.

Definition 0.14. A generator matrix for the dual code $C^\perp$ is called a *control matrix* for C .

We can check whether $\mathbf{x} \in C$ by checking if $\mathbf{x}H^T = \mathbf{0}$ where H is a control matrix. (Recall $(C^\perp)^\perp = C$.)

How can one find a control matrix?

Theorem 0.15. If a linear $[n, m]$ code C has the generator matrix $G = [I_m \mid A]$ then $H = [-A^T \mid I_{n-m}]$ is a control matrix.

Proof. (Sketch/Handout) Since $\dim C^\perp = n - \dim C = n - m$, the matrix H should be an $(n - m) \times n$ matrix. A matrix $\tilde{H}$ of this size with linearly independent rows satisfying $G\tilde{H}^T = 0$ has $n - m$ linearly independent rows in $C^\perp$ and hence must be a control matrix. Thus we just need to check that $H = [-A^T \mid I_{n-m}]$ is an $(n - m) \times n$ matrix and that $GH^T = 0$.

For the size aspect, consider $G = [I_m \mid A]$ and note that A is an $m \times (n - m)$ matrix. Therefore $H = [-A^T \mid I_{n-m}]$ is an $(n - m) \times n$ matrix, as required.

Lastly, note that

$$\begin{aligned} GH^T &= [I_m \mid A] \begin{bmatrix} -A \\ I_{n-m} \end{bmatrix} \\ &= -I_m A + A I_{n-m} \\ &= -A + A \\ &= 0, \end{aligned}$$

hence the result. $\square$

Example 0.16. (Handout) The code with $G = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix}$ in $\mathbb{Z}_3^4$ (the perfect code from before) has control matrix

$$H = \begin{pmatrix} 2 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix}.$$

0.3. Detection and correction of errors. We detect errors by computing $\mathbf{x}H^T$. If $\mathbf{x}H^T = \mathbf{0}$ then $\mathbf{x} \in C$, otherwise $\mathbf{x} \notin C$.

The row vector $\mathbf{x}H^T$ of length $n - m$ is called a *syndrome*. To correct $\mathbf{x}$ we look at all $\mathbf{y} \in F^n$ with the same syndrome as $\mathbf{x}$. The reason is that if $\mathbf{x}H^T = \mathbf{y}H^T$, then $(\mathbf{x} - \mathbf{y})H^T = \mathbf{0}$, and so $\mathbf{x} - \mathbf{y} \in C$. As we want to alter $\mathbf{x}$ as little as possible we search the $\mathbf{y}$ of lowest weight satisfying $\mathbf{y}H^T = \mathbf{x}H^T$. Such a $\mathbf{y}$ is called a *coset leader* (and the whole set $\{\mathbf{y} \in F^n \mid \mathbf{y}H^T = \mathbf{x}H^T\}$ is called a *coset*). We replace $\mathbf{x}$ by the code word $\mathbf{x} - \mathbf{y}$.

In the above example (the perfect code) we have

syndrome	00	01	02	10	11	12	20	21	22
coset leader	0000	0001	0002	0010	2000	0200	0020	0100	1000

To find the coset leaders we only need to look at H as all syndromes here are multiples of columns. (That is because every word is within distance one from a code word in this example.)

Now assume that we received the words:

$$\mathbf{x}_1 = 0110 \quad \mathbf{x}_2 = 0221 \quad \mathbf{x}_3 = 1111$$

We compute their syndromes using H , and we obtain 01, 00 and 21. This shows that 0221 is a code word, but not the other two. To correct $\mathbf{x}_1$ we subtract the corresponding coset leader 0001 to get $0110 - 0001 = 0112$. In the case of $\mathbf{x}_3$, the coset leader is 0100 and the corrected word is $1111 - 0100 = 1011$.

Remark 0.17. We observe that our general principle is to always assume that as few errors as possible have occurred. Consequently, our decoding strategy is to look for the nearest code word to the word which we have received.

Example 0.18. (Handout) It is not always the case that every coset has a unique coset leader. Consider for example $G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$ in $\mathbb{Z}_2^5$ and the related control matrix

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}. \text{ We have } 2^3 = 8 \text{ syndromes:}$$

$$000 \quad 001 \quad 010 \quad 011 \quad 100 \quad 101 \quad 110 \quad 111$$

The syndrome 000 corresponds to code words and five other syndromes correspond to columns of H and therefore have a unique coset leader (since all columns of H are different in this example). However, the syndromes 110 and 111 do not occur as columns so we check if they are a linear combination of two columns. They both are, but not in a unique way. For 110 is column 3 plus column 4, but also column 1 plus column 2. Therefore either (00110) or (11000) could be chosen as a coset leader. This means we do not really know how to correct words with these syndromes. (This tallies with Theorem 7.8 and the fact that the separation of the code is 3, as we will see below.)

0.4. Finding the separation from the control matrix.

Theorem 0.19. *A linear code C with control matrix H has separation σ if and only if there are σ linearly dependent columns in H but any $\sigma - 1$ columns are linearly independent.*

Proof. Every code word $\mathbf{x}$ satisfies $\mathbf{x}H^T = \mathbf{0}$ if and only if $H\mathbf{x}^T = \mathbf{0}$ and hence corresponds to the coefficients of a linear dependence of the columns of H . If $\mathbf{x}$ is a code word of minimal weight σ then exactly σ of the coefficients in the dependence are non-zero. If we had a dependence between $\sigma - 1$ columns this would correspond to a code word of weight $\leq \sigma - 1$, a contradiction. $\square$

Example 0.20. (Handout) In the above example with

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

columns 2, 4 and 5 are dependent: $c_2 - c_4 - c_5 = \mathbf{0}$. (This corresponds to $\mathbf{x} = (0, 1, 0, 1, 1) \in C$.) However all pairs of columns are independent. This shows that the separation is 3 in this case.